

Number of Functional Features Included in the Solution

CrediShield AI — Credit Card Approval Prediction System

Complete Feature Inventory

The final solution incorporates 20 distinct functional features spanning Machine Learning, Backend architecture, Frontend UI/UX, and DevOps deployment.

1. Machine Learning & Analytics Features (6 Features)

#	Feature Name	Description	Benefit
1	Automated Binary Classification	Ingests applicant data and predicts Approvable (1) or Rejectable (0) using a trained Gradient Boosting Classifier.	Replaces manual review with a fast, data-driven decision.
2	Confidence Scoring	Calculates a precise probability score (0% to 100%) for every prediction utilizing the model's <code>predict_proba()</code> method.	Quantifies the certainty behind the AI's recommendation.
3	Risk Level Grading	Automatically assigns a Risk Level tag (Low, Medium, or High) based on predefined probability thresholds.	Provides an easily understandable risk summary for analysts.
4	Multi-Model Comparison Engine	Trains 6 different ML algorithms (Logistic Regression, Trees, Forests, Boosting variants) simultaneously and records their performance metrics.	Ensures the system is evaluating the problem through multiple mathematical approaches.
5	Auto-Selection Protocol	Automatically evaluates all trained models and selects the "winner" based on the highest F1-Score.	Guarantees the deployed model is optimally tuned for handling the class imbalance.
6	Feature Importance Extraction	Analyzes the winning model's internal structure to identify and rank which applicant details (e.g., Income vs. Age) most influence the decision.	Enhances model explainability and transparency.

2. Frontend User Interface Features (8 Features)

#	Feature Name	Description	Benefit
7	Interactive Prediction Form	A structured, 14-field data entry form with pre-populated dropdown menus and numerical inputs.	standardizes data entry and minimizes typos.
8	Animated SVG Confidence Gauge	A circular progressing ring that dynamically fills with a gradient (Green for Approved, Red for Rejected) synced to a numeric count-up animation.	Highly engaging visual feedback summarizing the prediction result.
9	Interactive Visual Analytics	Four dynamic dashboards built with Chart.js displaying feature importance bars, approval ratio doughnuts, and demographic distributions.	Replaces static Excel reporting with real-time portfolio insights.
10	Prediction History Log	An interactive table that automatically records past predictions locally, displaying transaction IDs, timestamps, and outcomes.	Provides an instant audit trail without requiring a backend database.
11	Printable PDF Assessment Reports	Generates a branded "CrediShield AI Credit Assessment Letter" containing all applicant metadata and the final decision, formatted specifically for printing.	Fulfills regulatory compliance and documentation requirements with a single click.
12	Tabbed Dashboard Navigation	A single-page application (SPA) architecture allowing seamless switching between Predict, Metrics, Analytics, and History views without page reloads.	Improves application speed and user experience.
13	Obsidian Glassmorphism Theme	A premium design system utilizing deep-space background colors, frosted glass panels (<code>backdrop-filter: blur</code>), and glowing borders.	Delivers a modern, high-end aesthetic far superior to generic banking interfaces.
14	Persistent Light/Dark Mode Toggle	Allows users to swap the entire UI theme via a header button, with the preference saved to <code>localStorage</code> across sessions.	Enhances accessibility and user comfort in different lighting environments.

3. Backend Architecture Features (4 Features)

#	Feature Name	Description	Benefit
15	JSON REST API Endpoint	The <code>/predict</code> endpoint accepts a JSON payload of applicant details and returns a structured JSON object containing the decision.	Enables integration with other banking systems or mobile apps in the future.
16	Metrics Data Endpoint	The <code>/metrics</code> endpoint serves cached model evaluation scores and demographic statistics to the frontend.	Decouples heavy data processing from UI rendering, keeping charts fast.
17	Self-Healing Auto-Training	The server checks for required model artifacts on startup. If missing, it automatically triggers the <code>train.py</code> script to regenerate them.	Simplifies setup for new developers by removing manual initialization steps.
18	Robust Input Validation	The API validates the presence of all 14 required fields and traps encoding errors, returning descriptive HTTP 400 Bad Request messages.	Prevents server crashes from bad data and provides clear feedback to the UI.

4. DevOps & Deployment Features (2 Features)

#	Feature Name	Description	Benefit
19	Cloud Auto-Deployment	Fully configured to host on Render.com using Gunicorn, with a GitHub webhook triggering automatic rebuilds upon code pushes.	Ensures the application is always accessible globally without manual server administration.
20	Version-Pinned Environments	Dependencies are strictly separated between development (<code>requirements-train.txt</code>) and production (<code>requirements.txt</code>), with exact versions pinned (e.g., <code>scikit-learn==1.5.0</code>).	Solves cloud bundle-size limits and prevents catastrophic deserialization errors in production.

Feature Summary

- **Total Features:** 20
- **Development Status:** 100% Implemented
- **Integration:** Fully integrated into a cohesive, end-to-end application.

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